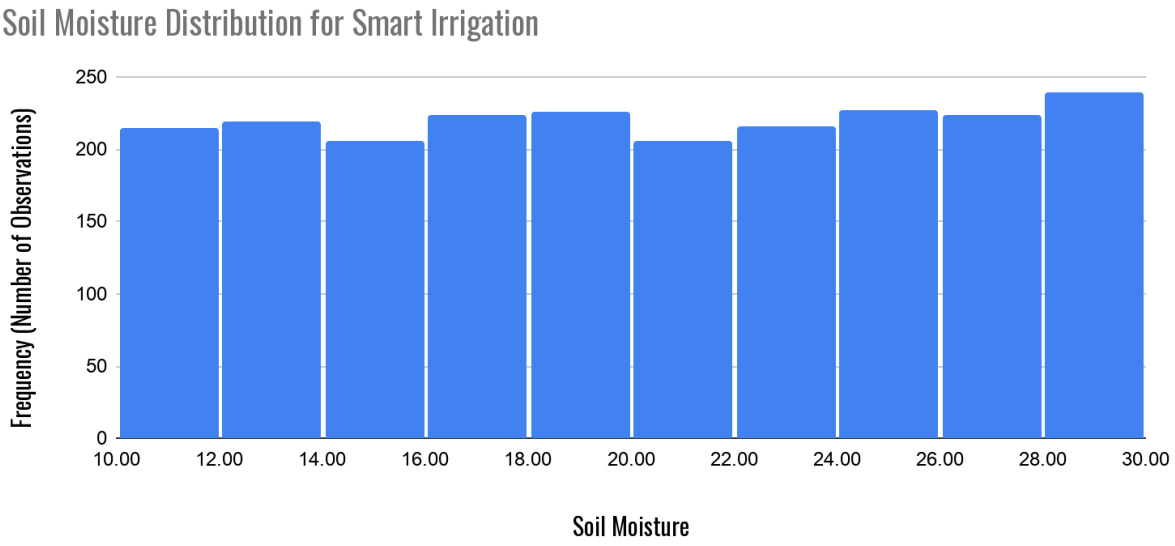
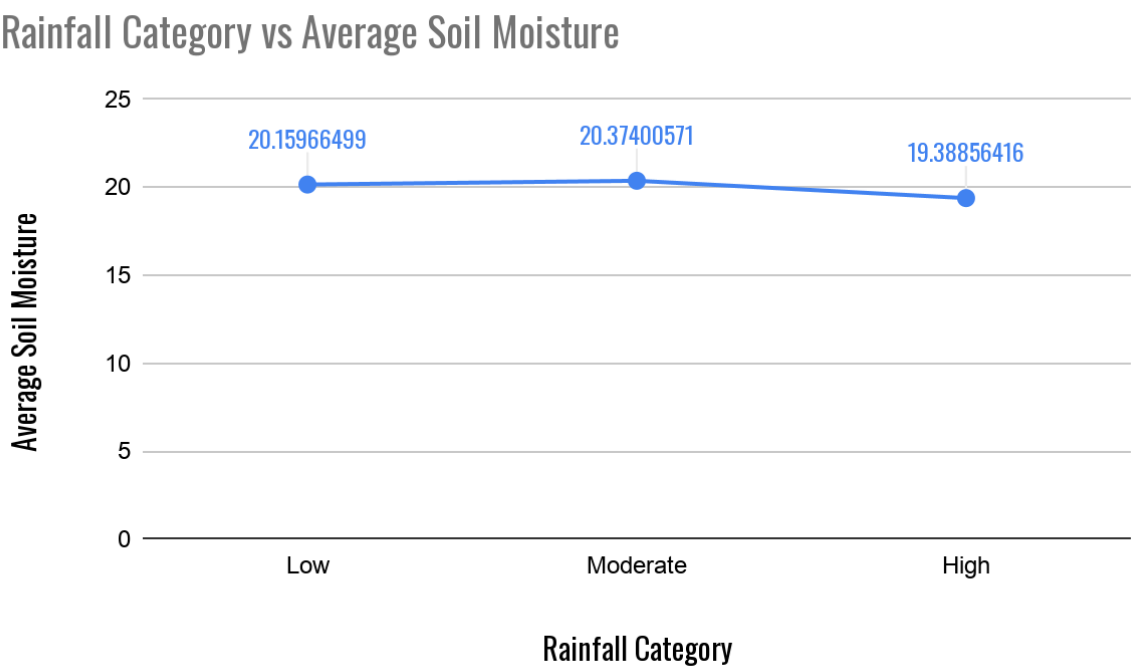
SMP IA GRAPHS

1. HISTOGRAM:



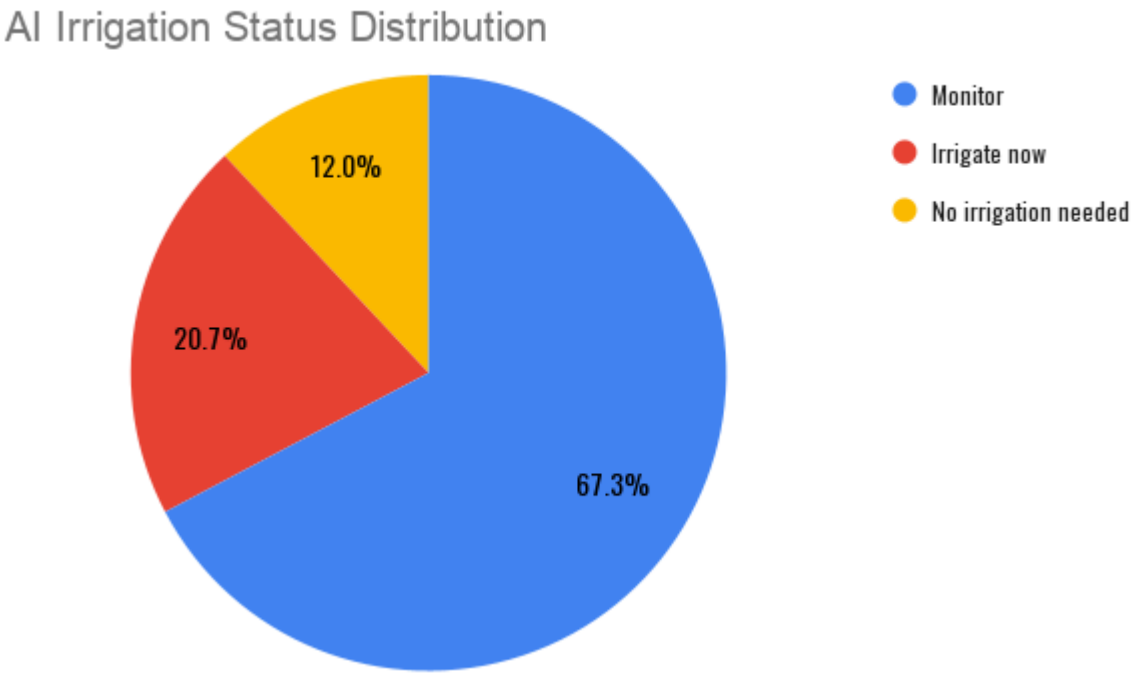
Interpretation: The histogram represents the frequency distribution of soil moisture values for 2,200 observations. The mean soil moisture is approximately 20.15, while the standard deviation is 5.79, indicating variation in the observed soil moisture levels. This analysis helps understand soil conditions for smart irrigation planning.

2. LINE GRAPH:



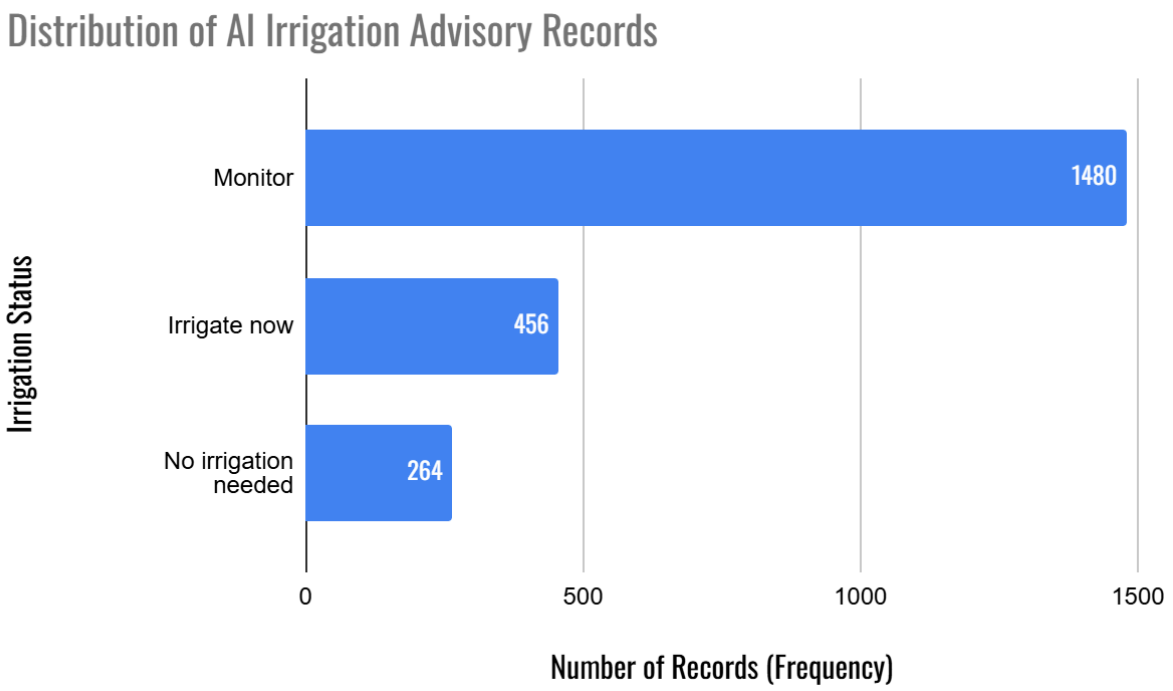
Interpretation: The line graph compares average soil moisture across low, moderate, and high rainfall categories. It helps examine how soil moisture varies under different rainfall conditions. This information can support irrigation planning, although rainfall category alone cannot determine the irrigation requirement.

3. PIE CHART:



Interpretation: The pie chart shows the distribution of 2,200 irrigation advisory records. Monitor accounts for 67.27%, Irrigate Now for 20.73%, and No Irrigation Needed for 12% of the records. This distribution summarises the irrigation advisory outcomes and can help in understanding irrigation planning requirements.

4. BAR GRAPH:



Interpretation: The bar graph compares the frequency of three AI irrigation advisory statuses across 2,200 records. Monitor has the highest frequency with 1,480 records, followed by Irrigate Now with 456 records and No Irrigation Needed with 264 records. This comparison helps summarise the distribution of irrigation advisory outcomes.